Bioinformatics Training

Session 1

Nigerian Liver Cancer Molecular Research Group

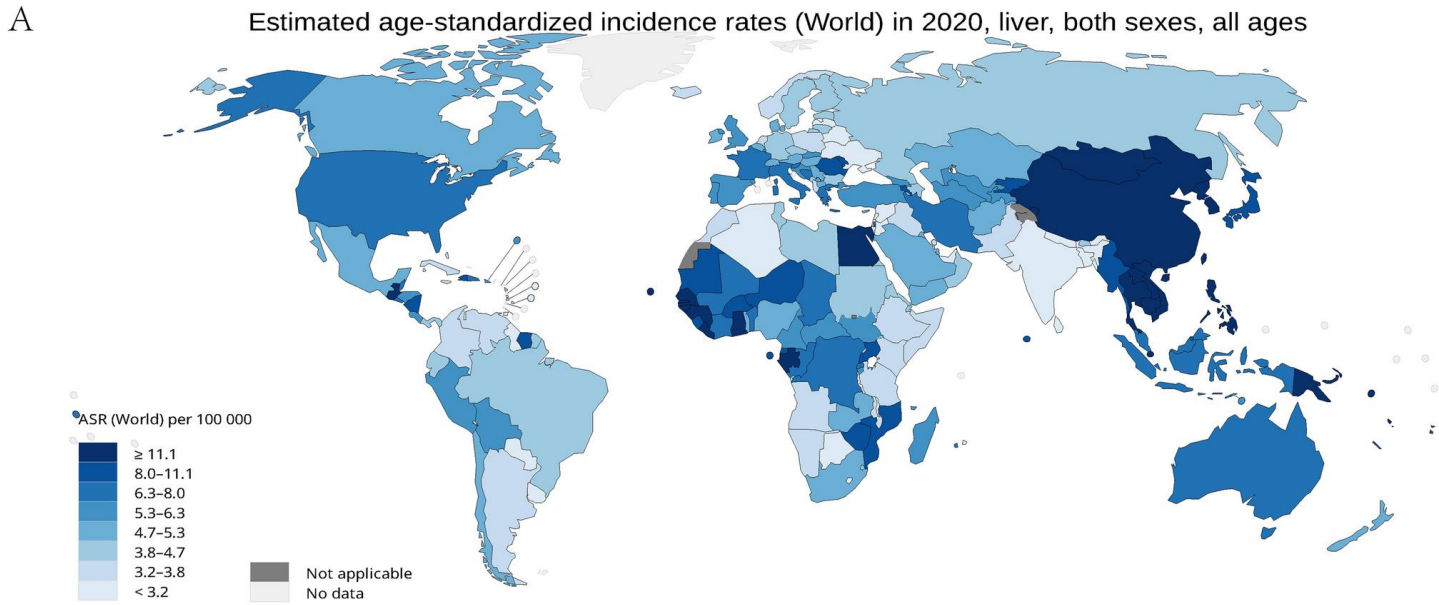
Olabode Omotoso

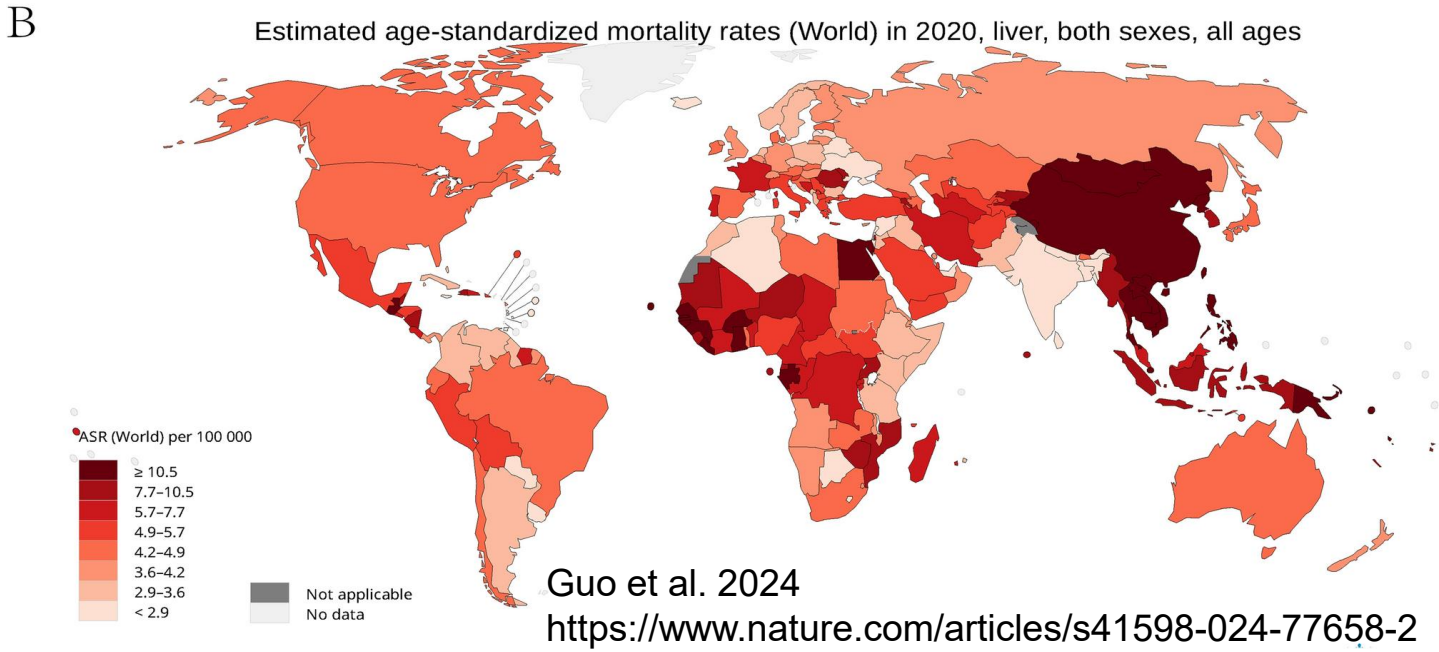
October 4, 2026

Why are we here?

Challenges in liver cancer research

- High burden in Africa
- Genetic susceptibility
- Environmental risk factors
- Need for African genomic data





Guo et al. 2024
https://www.nature.com/articles/s41598-024-77658-2

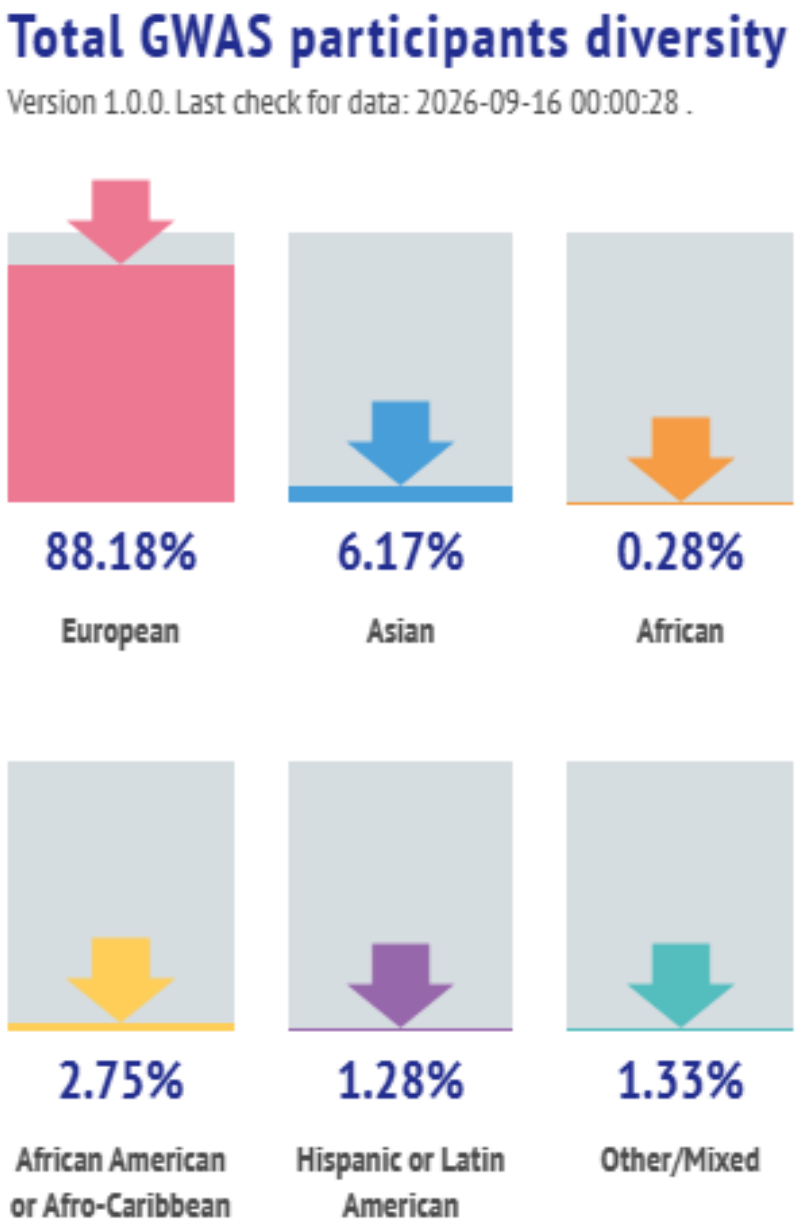
Why are we here?

What is Bioinformatics?

Application of computational methods to biological data

Bioinformatics help us

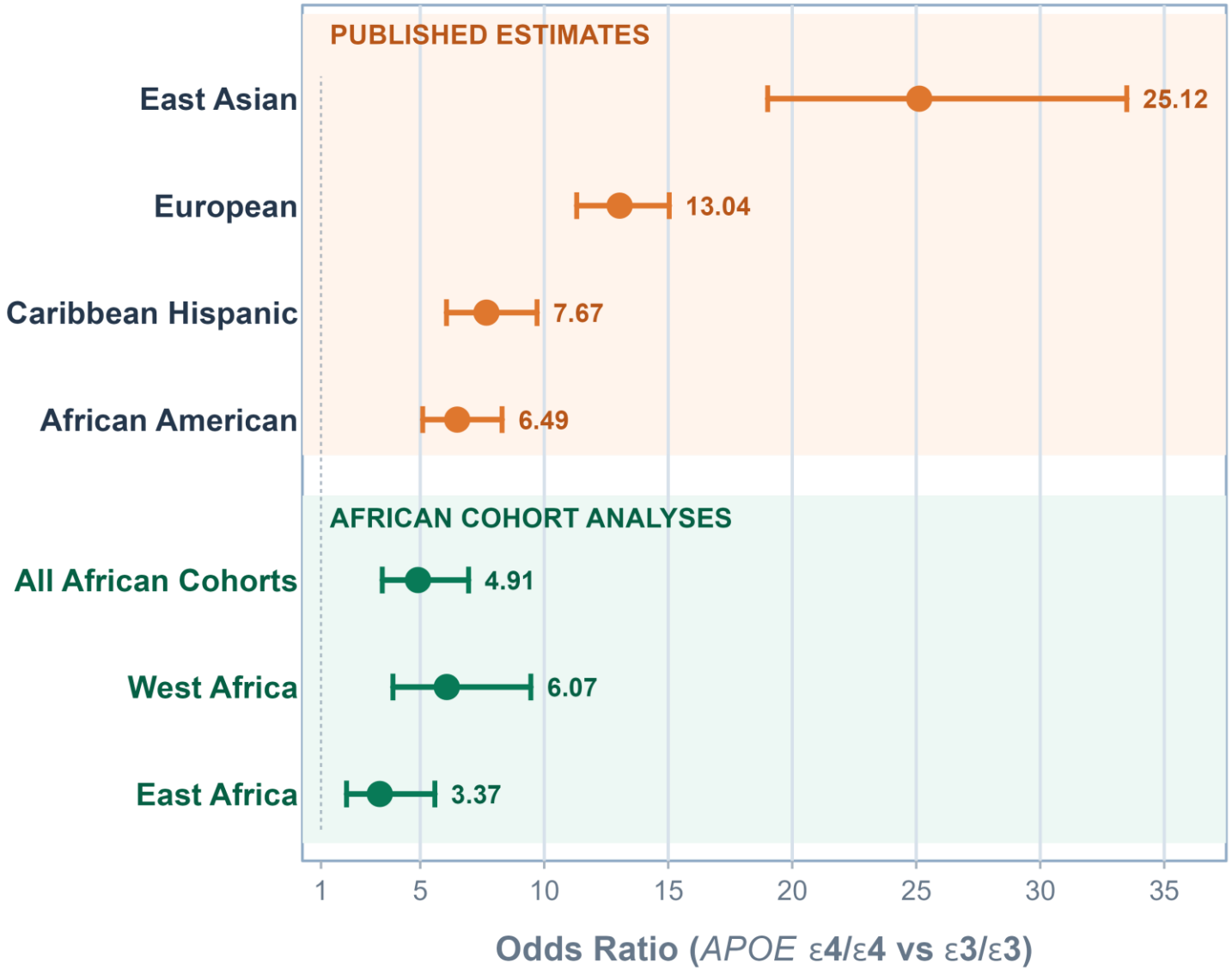
- Process genomic data
- Identify risk and protective variants
- Understand ancestry
- Discover biomarkers, mechanisms or pathways



https://gwasdiversitymonitor.com/



APOE ε4/ε4 Risk Vary Substantially Across Populations

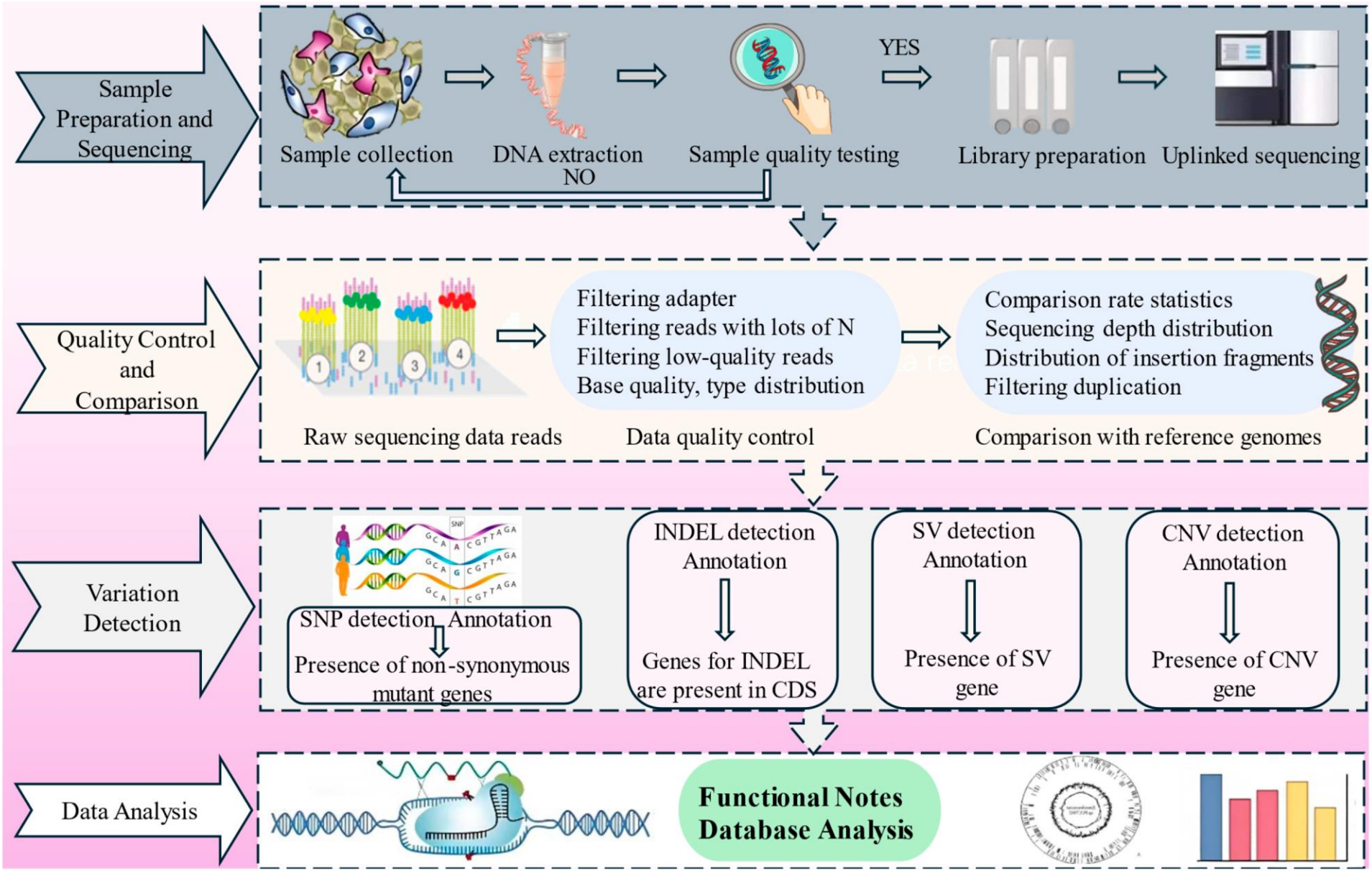


ASIAN ε4/ε4 OR=25.12 → **AFRICAN ε4/ε4 OR=3.37**

Same variant: ~7x lower risk

What's protecting **East Africans** could be the next AD drug target and only now visible because we studied **indigenous Africans.**

Typical Genomics Workflow



Lu et al. 2025 https://www.mdpi.com/1422-0067/26/1/372#

Why we must learn Linux

Most genomics software runs on Linux:

- PLINK
- BCFtools
- Samtools
- ADMIXTURE
- GATK
- SAIGE or REGENIE

Training run down

- Command line
- R programming
- Population genetics concepts
- GWAS concepts – QC, PCA, Admixture
- Post-GWAS and interpretation

Command Line Foundations 1

- **Concepts:** Terminal vs GUI, filesystem, paths
- **Hands-on:** pwd, ls, cd, mkdir, cp, mv, rm, man
- **Focus:** folder structure, cheat sheet and wild cards

Command Line Foundations 2

- **Concepts:** piping, redirection, creating and viewing files (vi, nano, touch)
- **Hands-on:** grep, cut, sort, uniq, awk
- **Focus:** preparing and navigating files

Command Line Foundations I

GUI vs Command Line

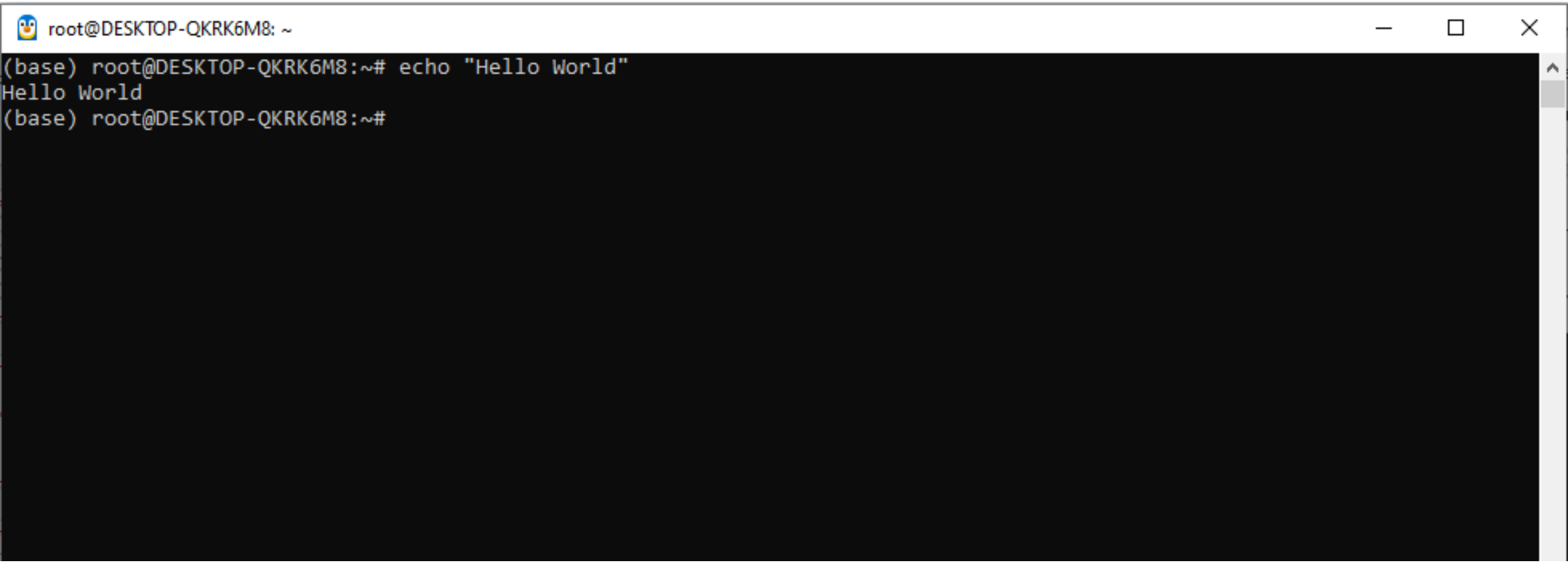
GUI

- Mouse Based
- Easier initially
- Limited automation
- Difficult for large-scale analyses

Command Line

- Text-based
- More powerful
- Highly automated
- Ideal for big datasets

THE TERMINAL, SHELL and BASH



The same or different?

THE TERMINAL, SHELL and BASH



https://www.vecteezy.com/vector-art/12494902

Terminal
Window where we type commands

Shell
Interpreter that understands and executes the command

Bash
Most widely used Linux shell

The same or different?



redbubble.com



Summary of today's hands-on session

Recap:

- Intro. to Bioinformatics and liver cancer genomics
- history
- clear
- man
- ls (-lha)
- Using tab for auto-completion
- pwd
- cd (~, ../.., ..)
- mkdir
- chmod
- chmod (drwx, ugo, 777, 421)

Link to additional resource
https://en.wikipedia.org/wiki/Command-line_interface